

MA 301
Test 1
Spring - 2014

Problem	points	out of
1		5
2		5
3		5
4		5
5		10
6		8
Score		38

Name: _____

HR: _____

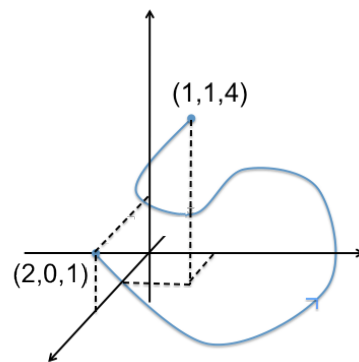
Signature: _____

READ BEFORE YOU START
* Show all work. * Change your calculator mode to RADIANS. * Sloppy solutions may be penalized. * Use calculators ONLY for computations. * You must write up every problem in a clear and organized way. * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Let $\mathbf{F} = [z^2, x - y, 7]$ and C be the intersection curve of the surfaces $x = y + z$ and $y = z^2$.

Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$.

Problem 2: Let $\mathbf{F} = \left[\frac{y^2}{x}, 2y \ln x, 3z^2 \right]$ compute the work required to move an object along C (shown below). Justify your work.



Problem 3: Let C be the triangle with vertices $(0,0)$, $(1,0)$, $(1,4)$ oriented counter-clockwise and

$$\mathbf{F}(x,y) = (xe^{y^2} - 3xy)\mathbf{i} + \frac{3}{2}y^2\mathbf{j}$$

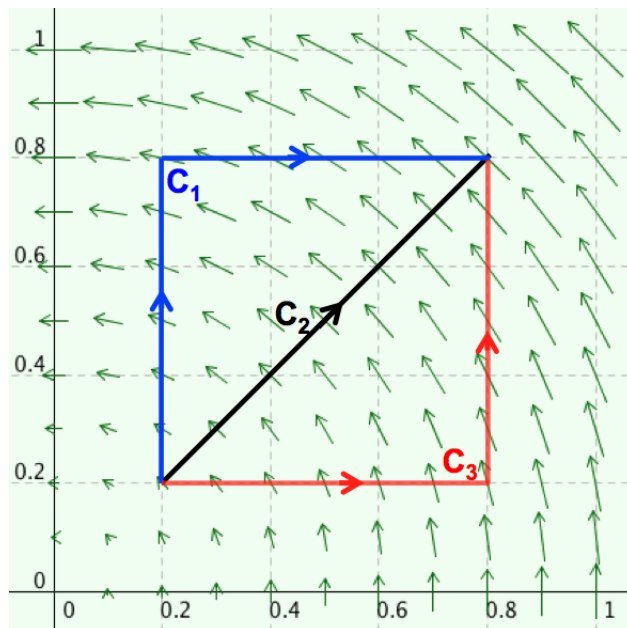
Find the flux of $\mathbf{F}$ around C : $\int_C \mathbf{F} \cdot \mathbf{N} \, ds$ where $\mathbf{N}$ is the unit normal to C

Problem 4: Let R be a region bounded by the upper half of the circle $x^2 + y^2 = 10$ and the x-axis. Let C be the boundary of R with counter-clockwise orientation. Compute the following line integral

$$\int_C (15x - 4xy) \, dx + (\cos x - 6xy) \, dy$$

Problem 5: Answer the following questions.

(a) Consider the vector field $\mathbf{F}$ shown in the figure, together with the paths C_1 , C_2 , and C_3 . Order the line integrals $\left(\int_{C_1} \mathbf{F} \cdot d\mathbf{r}, \int_{C_2} \mathbf{F} \cdot d\mathbf{r}, \int_{C_3} \mathbf{F} \cdot d\mathbf{r} \right)$ from smallest to largest. Explain your reasoning.



(b) Suppose $\mathbf{F}$ is conservative and $\int_{(0,0)}^{(0,1)} \mathbf{F} \cdot d\mathbf{r} = 2.4$, $\int_{(0,1)}^{(1,1)} \mathbf{F} \cdot d\mathbf{r} = -1.2$ and $\int_{(0,0)}^{(1,0)} \mathbf{F} \cdot d\mathbf{r} = 7.5$. Then

$$\int_{(1,0)}^{(1,1)} \mathbf{F} \cdot d\mathbf{r} =$$

(c) Let $\mathbf{F} = [-y, x]$ and C be a simple closed curve in the xy -plane.

Show that $\oint_C \mathbf{F} \cdot d\mathbf{r} = 2 \cdot \text{Area}(R)$ where R is the region bounded by C .

Problem 6: Let $\mathbf{F} = [x + y, 2xy]$ and C be the boundary of the region enclosed by the curves $y = x^2$, $y = x + 2$, and $x = 0$ in the first quadrant.

(a) Setup the integral you would use to compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ directly (*don't leave integrand as a dot product*)

(b) Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ using green's theorem.